Chpater1 Mechanics & Fluids

Lesson3 Work, Kinetic Energy and Power

1. A weight lifter lifts a 350-N set of weights from ground level to a position over his head, a vertical distance of 2.00 m. How much work does the weight lifter do, assuming he moves the weights at constant speed?
2. In 1990 Walter Arfeuille of Belgium lifted a 281.5-kg object through a distance of 17.1 cm using only his teeth. **(a)** How much work did Arfeuille do on the object? **(b)** What magnitude force did he exert on the object during the lift, assuming the force was constant?
3. A shopper in a supermarket pushes a cart with a force of 35 N directed at an angle of 25° below the horizontal. The force is just sufficient to overcome various frictional forces, so the cart moves at constant speed. **(a)** Find the work done by the shopper as she moves down a 50.0-m length aisle. **(b)** What is the net work done on the cart?
4. Starting from rest, a 5.00-kg block slides 2.50 m down a rough 30.0° incline. The coefficient of kinetic friction between the block and the incline is $\mu_k = 0.436$. Determine **(a)** the work done by the force of gravity, **(b)** the work done by the friction force between block and incline, and **(c)** the work done by the normal force.
5. A horizontal force of 150 N is used to push a 40.0-kg packing crate a distance of 6.00 m on a rough horizontal surface. If the crate moves at constant speed, find **(a)** the work done by the 150-N force and **(b)** the coefficient of kinetic friction between the crate and surface.
6. A sledge loaded with bricks has a total mass of 18.0 kg and is pulled at constant speed by a rope inclined at 20.0° above the horizontal. The sledge moves a distance of 20.0 m on a horizontal surface. The coefficient of kinetic friction between the sledge and surface is 0.500. **(a)** What is the tension in the rope? **(b)** How much work is done by the rope on the sledge? **(c)** What is the mechanical energy lost due to friction?
7. A block of mass $m = 2.50$ kg is pushed a distance $d = 2.20$ m along a frictionless horizontal table by a constant applied force of magnitude $F = 16.0$ N directed at an angle $\theta = 25.0°$ below the horizontal as shown in **Figure P5.8**. Determine the work done by **(a)** the applied force, **(b)** the normal force exerted by the table, **(c)** the force of gravity, and **(d)** the net force on the block.

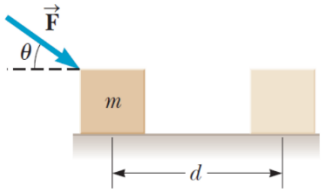


Figure P5.8

8. A mechanic pushes a 2.50×10^3-kg car from rest to a speed of v, doing 5 000 J of work in the process. During this time, the car moves 25.0 m. Neglecting friction between car and road, find **(a)** v and **(b)** the horizontal force exerted on the car.
9. A 7.00-kg bowling ball moves at 3.00 m/s. How fast must a 2.45-g Ping-Pong ball move so that the two balls have the same kinetic energy?

10. A 65.0-kg runner has a speed of $5.20\ \mathrm{m/s}$ at one instant during a long-distance event. **(a)** What is the runner's kinetic energy at this instant? **(b)** If he doubles his speed to reach the finish line, by what factor does his kinetic energy change?

11. A worker pushing a 35.0-kg wooden crate at a constant speed for $12.0\ \mathrm{m}$ along a wood floor does $350\ \mathrm{J}$ of work by applying a constant horizontal force of magnitude F_0 on the crate. **(a)** Determine the value of F_0. **(b)** If the worker now applies a force greater than F_0, describe the subsequent motion of the crate. **(c)** Describe what would happen to the crate if the applied force is less than F_0.

12. A 70-kg base runner begins his slide into second base when he is moving at a speed of $4.0\ \mathrm{m/s}$. The coefficient of friction between his clothes and Earth is 0.70. He slides so that his speed is zero just as he reaches the base. **(a)** How much mechanical energy is lost due to friction acting on the runner? **(b)** How far does he slide?

13. A running 62-kg cheetah has a top speed of $32\ \mathrm{m/s}$. **(a)** What is the cheetah's maximum kinetic energy? **(b)** Find the cheetah's speed when its kinetic energy is one half of the value found in part **(a).**

14. A 7.80-g bullet moving at $575\ \mathrm{m/s}$ penetrates a tree trunk to a depth of $5.50\ \mathrm{cm}$. **(a)** Use work and energy considerations to find the average frictional force that stops the bullet. **(b)** Assuming the frictional force is constant, determine how much time elapses between the moment the bullet enters the tree and the moment it stops moving.

15. A 0.60-kg particle has a speed of $2.0\ \mathrm{m/s}$ at point A and a kinetic energy of $7.5\ \mathrm{J}$ at point B. What is **(a)** its kinetic energy at A? **(b)** Its speed at point B? **(c)** The total work done on the particle as it moves from A to B?

16. A large cruise ship of mass $6.50 \times 10^7\ \mathrm{kg}$ has a speed of $12.0\ \mathrm{m/s}$ at some instant. **(a)** What is the ship's kinetic energy at this time? **(b)** How much work is required to stop it? **(c)** What is the magnitude of the constant force required to stop it as it undergoes a displacement of $2.50\ \mathrm{km}$?

17. A man pushing a crate of mass $m = 92.0\ \mathrm{kg}$ at a speed of $v = 0.850\ \mathrm{m/s}$ encounters a rough horizontal surface of length $l = 0.65\ \mathrm{m}$ as in **Figure P5.18**. If the coefficient of kinetic friction between the crate and rough surface is 0.358 and he exerts a constant horizontal force of $275\ \mathrm{N}$ on the crate, find **(a)** the magnitude and direction of the net force on the crate while it is on the rough surface, **(b)** the net work done on the crate while it is on the rough surface, and **(c)** the speed of the crate when it reaches the end of the rough surface.

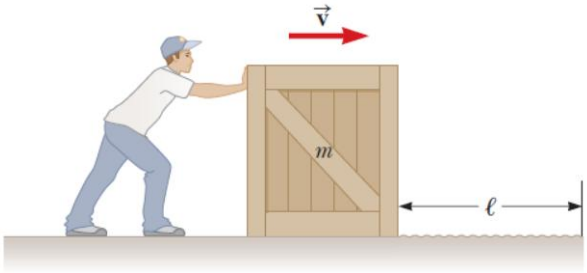


Figure P5.18

18. A 0.20-kg stone is held $1.3\ \text{m}$ above the top edge of a water well and then dropped into it. The well has a depth of $5.0\ \text{m}$. Taking $y = 0$ at the top edge of the well, what is the gravitational potential energy of the stone–Earth system **(a)** before the stone is released and **(b)** when it reaches the bottom of the well. **(c)** What is the change in gravitational potential energy of the system from release to reaching the bottom of the well?

19. When a 2.50-kg object is hung vertically on a certain light spring described by Hooke's law, the spring stretches $2.76\ \text{cm}$. **(a)** What is the force constant of the spring? **(b)** If the 2.50-kg object is removed, how far will the spring stretch if a 1.25-kg block is hung on it? **(c)** How much work must an external agent do to stretch the same spring $8.00\ \text{cm}$ from its unstretched position?

20. A daredevil on a motorcycle leaves the end of a ramp with a speed of $35.0\ \text{m/s}$ as in **Figure P5.25**. If his speed is $33.0\ \text{m/s}$ when he reaches the peak of the path, what is the maximum height that he reaches? Ignore friction and air resistance.

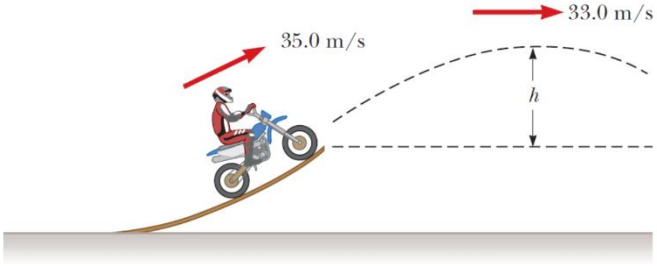


Figure P5.25

21. A 50.0-kg projectile is fired at an angle of $30.0°$ above the horizontal with an initial speed of $1.20 \times 10^2\ \text{m/s}$ from the top of a cliff $142\ \text{m}$ above level ground, where the ground is taken to be $y = 0$. **(a)** What is the initial total mechanical energy of the projectile? **(b)** Suppose the projectile is traveling $85.0\ \text{m/s}$ at its maximum height of $y = 427\ \text{m}$. How much work has been done on the projectile by air friction? **(c)** What is the speed of the projectile immediately before it hits the ground if air friction does one and a half times as much work on the projectile when it is going down as it did when it was going up?

22. A projectile of mass m is fired horizontally with an initial speed of v_0 from a height of h above a flat, desert surface. Neglecting air friction, at the instant before the projectile hits the ground, find the following in terms of m, v_0, h, and g: **(a)** the work done by the force of gravity on the projectile, **(b)** the change in kinetic energy of the projectile since it was fired, and **(c)** the final kinetic energy of the projectile.

23. A child and a sled with a combined mass of $50.0\ \text{kg}$ slide down a frictionless slope. If the sled starts from rest and has a speed of $3.00\ \text{m/s}$ at the bottom, what is the height of the hill?

24. Hooke's law describes a certain light spring of unstretched length $35.0\ \text{cm}$. When one end is attached to the top of a door frame and a 7.50-kg object is hung from the other end, the length of the spring is $41.5\ \text{cm}$. **(a)** Find its spring constant. **(b)** The load and the spring are taken down. Two people pull in opposite directions on the ends of the spring, each with a force of $190\ \text{N}$. Find the length of the spring in this situation.

25. A block of mass $m = 5.00\ \text{kg}$ is released from rest from point A and slides on the frictionless track shown in **Figure P5.36**. Determine **(a)** the block's speed at points B and C and **(b)** the net work done by the gravitational force on the block as it moves from point from A to C.

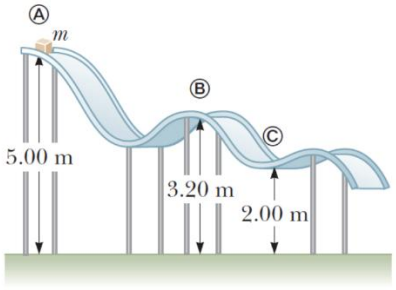


Figure P5.36

26. Tarzan swings on a 30.0-m-long vine initially inclined at an angle of $37.0°$ with the vertical. What is his speed at the bottom of the swing **(a)** if he starts from rest? **(b)** If he pushes off with a speed of $4.00\ \text{m/s}$?

27. Two blocks are connected by a light string that passes over a frictionless pulley as in **Figure P5.38**. The system is released from rest while m_2 is on the floor and m_1 is a distance h above the floor. **(a)** Assuming $m_1 > m_2$, find an expression for the speed of m_1 just as it reaches the floor. **(b)** Taking $m_1 = 6.5\ \text{kg}$, $m_2 = 4.2\ \text{kg}$, and $h = 3.2\ \text{m}$, evaluate your answer to part **(a),** and **(c)** find the speed of each block when m_1 has fallen a distance of $1.6\ \text{m}$.

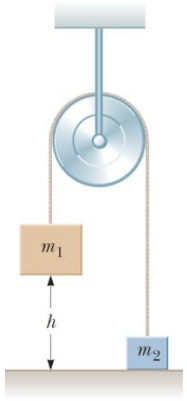


Figure P5.38

28. A skier of mass $70\ \text{kg}$ is pulled up a slope by a motor driven cable. **(a)** How much work is required to pull him $60\ \text{m}$ up a $30°$ slope (assumed frictionless) at a constant speed of $2.0\ \text{m/s}$? **(b)** What power (expressed in hp) must a motor have to perform this task?

29. While running, a person dissipates about $0.60\ \text{J}$ of mechanical energy per step per kilogram of body mass. If a 60-kg person develops a power of $70\ \text{W}$ during a race, how fast is the person running? (Assume a running step is $1.5\ \text{m}$ long.)

30. The electric motor of a model train accelerates the train from rest to $0.620\ \text{m/s}$ in $21.0\ \text{ms}$. The total mass of the train is $875\ \text{g}$. Find the average power delivered to the train during its acceleration.